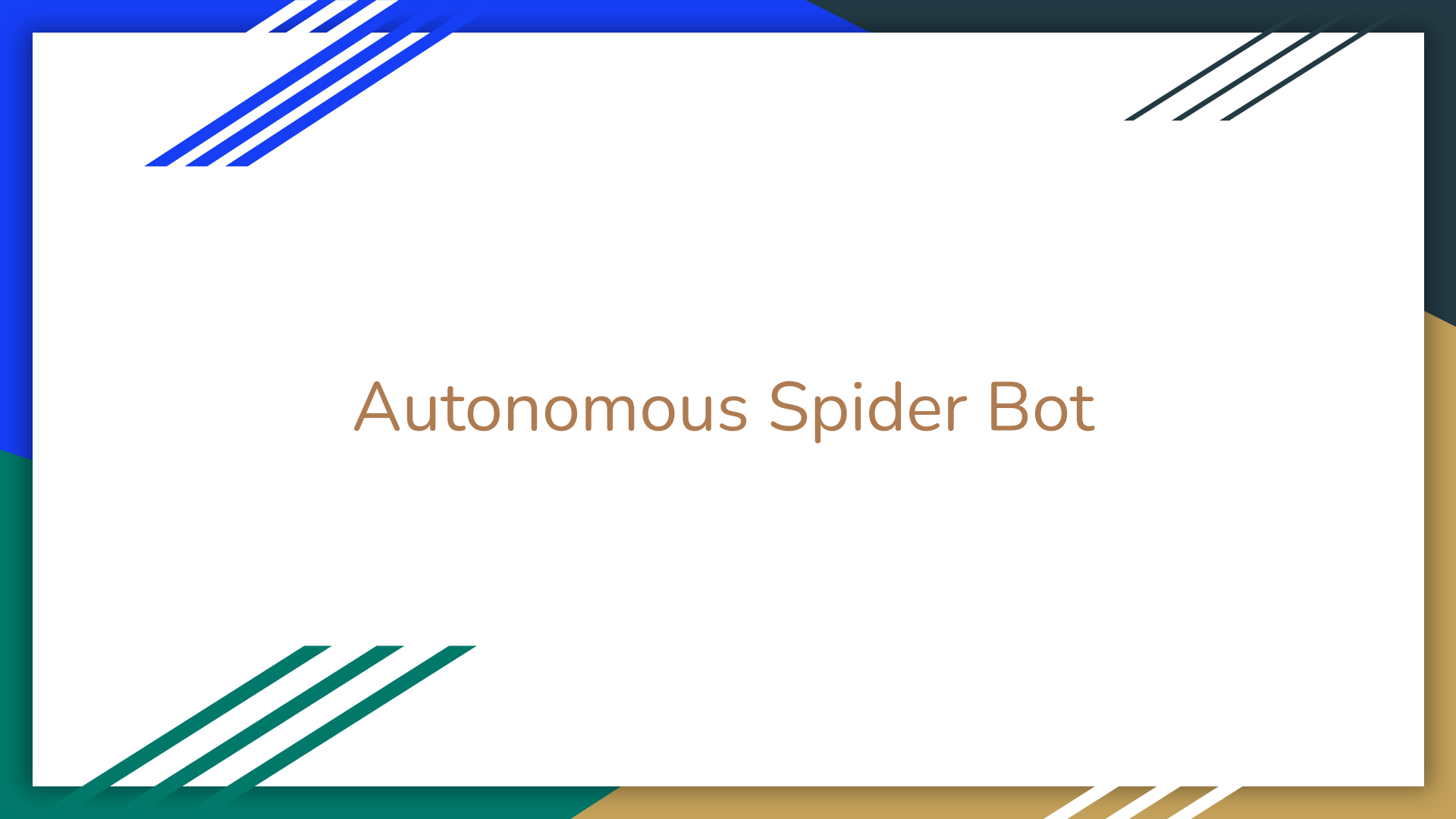




Autonomous Spider Bot

AI Project Introduction

The primary objective:

- To develop a **multi-legged autonomous** spider robot
- To utilize **reinforcement learning**
- To train model in a **simulation environment**
- To deploy the trained model on a **physical robot** (future goal)
- To enable future **autonomous navigation** (future goal)

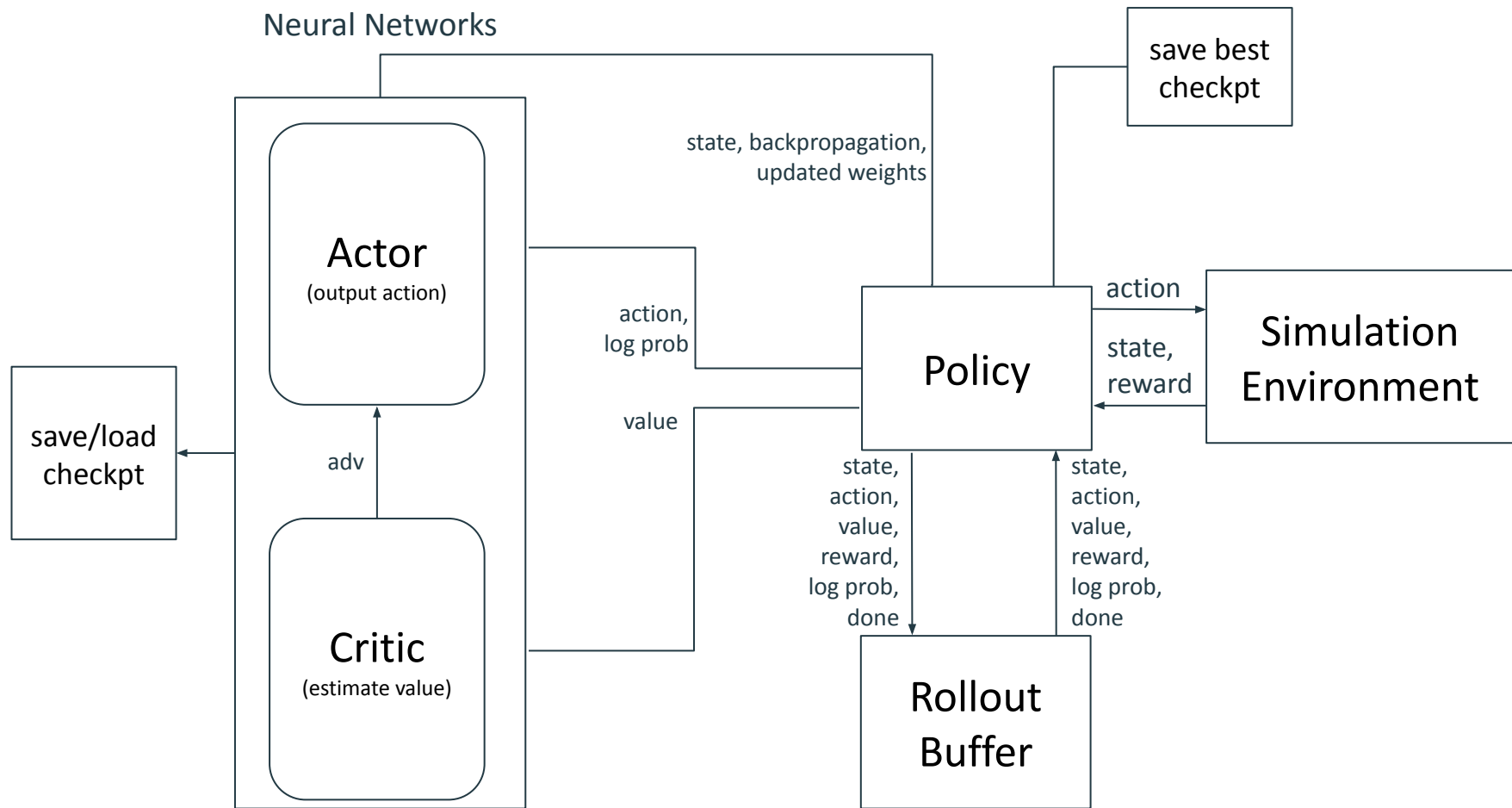
Problems & Project Motivation

Traditional Robot Control

- Relies on **manually designed** control rules
- **Performs poorly** in unseen or complex environments
- Requires **extensive computational resources**

Reinforcement Learning Approach

- Learns optimal behaviors via environment **feedback**
- Adapts to **complex** and **dynamic environments** in real time
- Enhances performance over time via **trial-and-error learning**



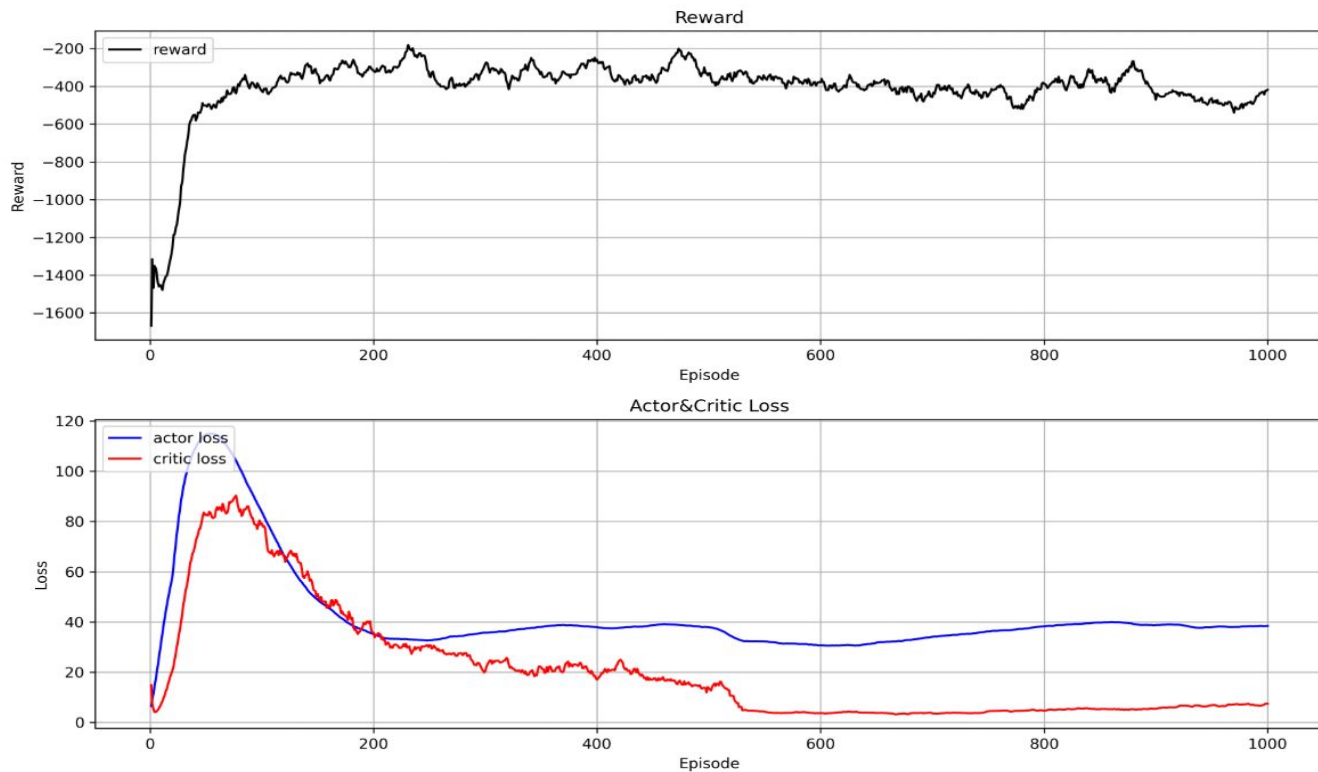
Key Differences between DDPG and PPO

	DDPG (Deep Deterministic Policy Gradient)	PPO (Proximal Policy Optimization)
Policy Type	Deterministic	Stochastic
Algorithm Type	Off-Policy	On-Policy
NN Method	Actor-Critic + Target Actor-Critic	Actor-Critic
Sample Efficiency	High	Low
Exploration	OU Noise	Policy Sampling + Entropy Bonus
Main Idea	DQN (Deep Q-Network) + Deterministic Policy Gradient (DPG)	Clipped Objective Function

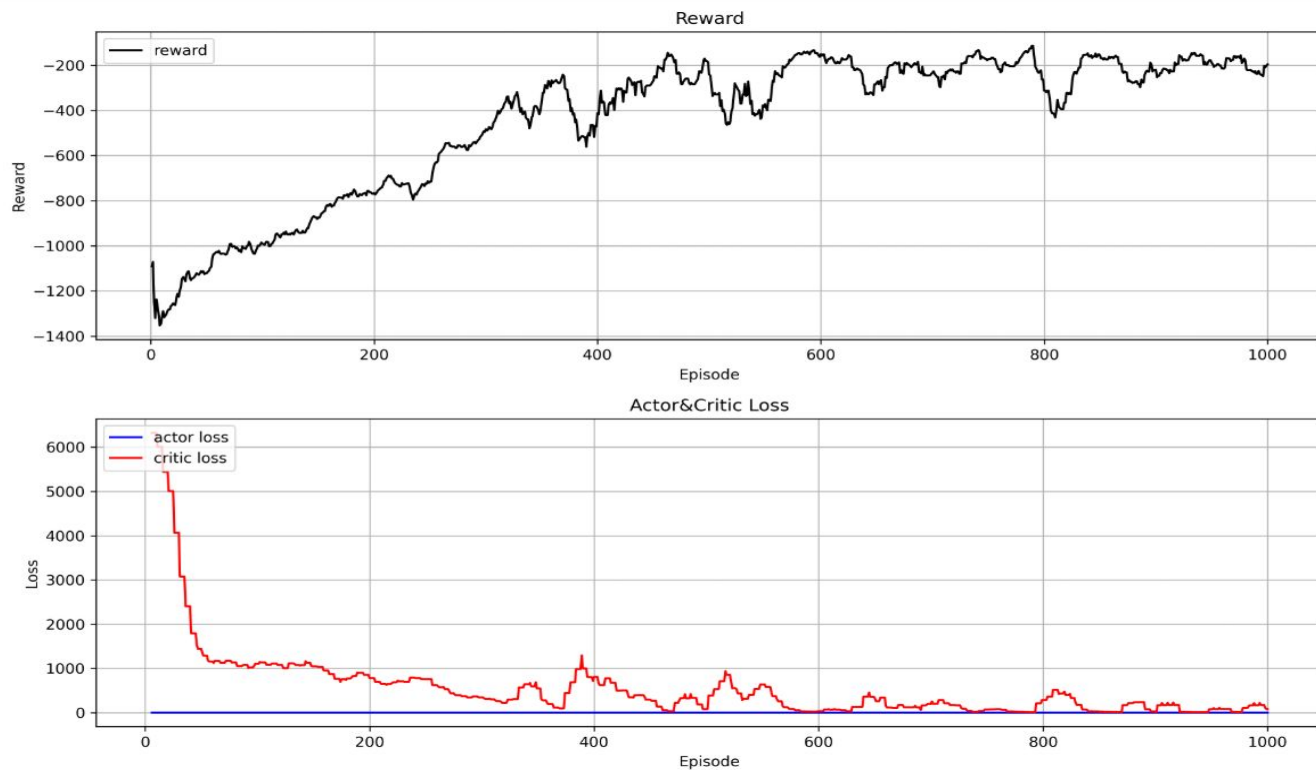
My Contributions

- **PPO Research & Implementation**
- **PPO vs DDPG** Performance Analysis
- **Pybullet-Based** Simulation Testing
- Reward & Loss **metrics** and Performance Visualization

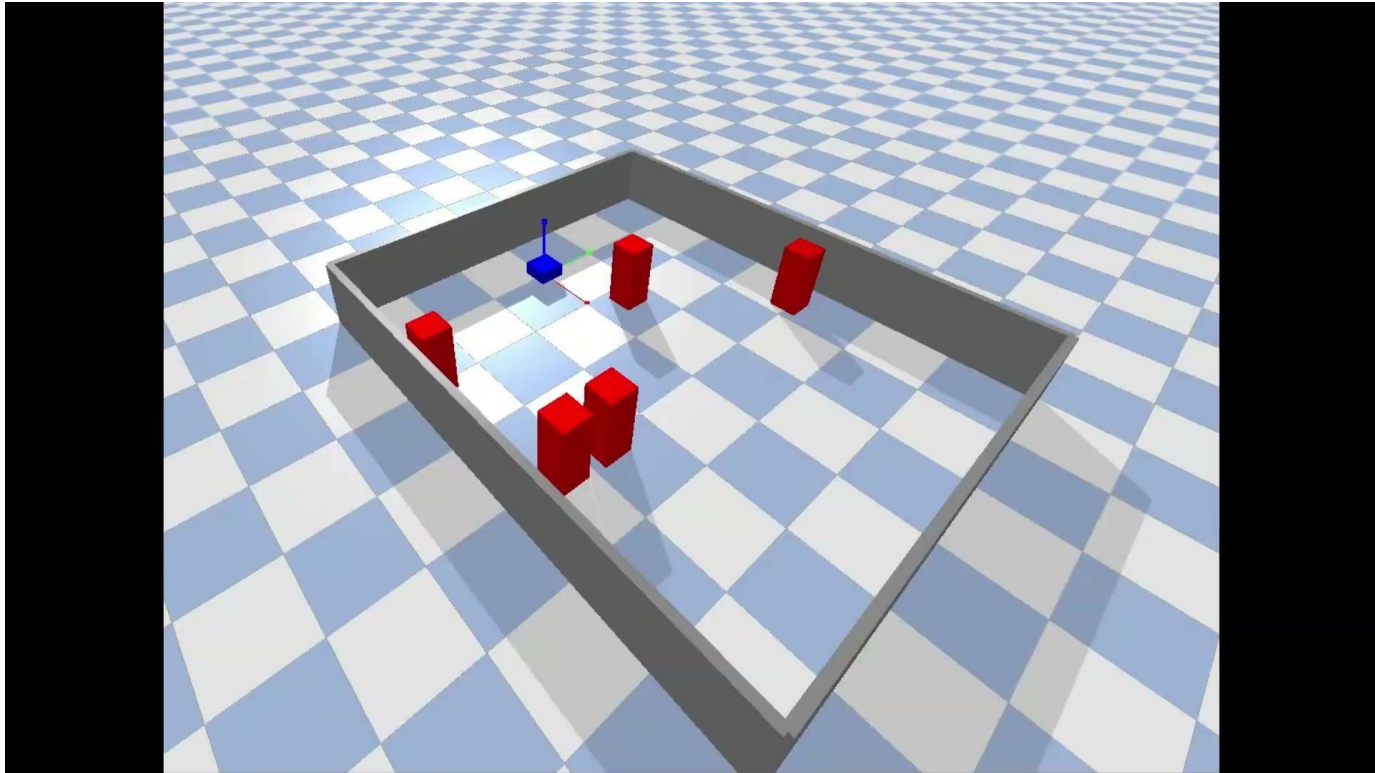
DDPG Training Performance in Pendulum-v1 Env



PPO Training Performance in Pendulum-v1 Env



Simulation Video (4x speed)



PPO Training Performance in Pybullet Env



PPO Training Performance in Pybullet Env

